Sejal Agarwal

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PROFESSIONAL SUMMARY

Software engineer and Master's Computer Science candidate at the University of Waterloo with 4+ years of experience building web and UI applications, including 2+ years of professional software engineering experience. Skilled in TypeScript, JavaScript, React/Next.js, and Angular, with a strong background in UI/UX, Human-Computer Interaction, and performant, maintainable front-end systems.

EDUCATION

Master of Mathematics (Computer Science), thesis-based
University of Waterloo — Ontario · Final CGPA 4.0/4.0 · Research in Human-Computer Interaction · September 2025 — Present

Bachelor of Software Engineering with AI specialization
University of Victoria — British Columbia · Final CGPA 4.0/4.0, with distinction. · September 2020 — April 2025

SKILLS

Frontend: TypeScript, JavaScript, React, Next.js, Angular, Tailwind CSS, HTML/CSS, responsive web development
Testing & APIs: Jest, REST APIs
Backend & Data: Node.js, npm, PostgreSQL, SQL, Supabase
Languages: Python, Java, Kotlin, C, MATLAB
Developer & Design Tools: Git, GitHub, Jira, Confluence, Figma, Linux/UNIX, macOS, Windows

EXPERIENCE

Software Engineer Intern
Outschool — January — April 2024, September — December 2024 | San Francisco, CA
- Increased new-buyer engagement/sign-ups by 25% by designing, implementing, and shipping A/B-tested UI/UX experiments across key workflows.
- Built and iterated on production web features using experimentation telemetry to evaluate performance and guide UI improvements.
- Collaborated with product and data teams to translate user and business needs into web experiences and data-driven product recommendations.

Software Engineer Co-op
Open Ocean Robotics — May — August 2023 | Victoria, BC
- Improved maintainability across a large-scale web codebase by migrating ~60,000 lines of code (40% of the repository) from JavaScript to TypeScript.
- Built a responsive web image carousel for live Unmanned Surface Vehicle feeds using APIs, caching, and lazy loading to optimize image delivery and reduce live-streaming costs by 20%.

Software Engineer Intern
Respondent — September — December 2022 | New York, NY
- Refactored ~20% of an Angular web application, removing legacy front-end code and shipping 5 new user-facing features to improve platform maintainability and stability.
- Raised coverage by writing Jest unit tests for new and existing code, reducing code smells before review.

Full-Stack Developer Co-op
19Labs — January — April 2022 | San Mateo, CA
- Drove a platform-wide user migration for the GALE telehealth platform, consolidating 7 permission roles into 4, re-mapping every user, and redesigning role-based UI and permissions for each user type.
- Integrated Bluetooth connectivity to stream real-time health metrics from medical devices (blood pressure monitors, glucometers), enabling healthcare practitioners to review live patient data on the platform.

PROJECTS

Otter: RAG Study Workspace + Student GenAI usage Dashboard for Instructors | 2026
- Built a responsive Next.js application with reusable UI components and role-based student/instructor experiences, backed by PostgreSQL & REST APIs.
- Engineered an instructor analytics dashboard with anonymized usage metrics and automated weekly summaries.

Martletplace: UVic Community Marketplace | 2025
- Led a 14-person engineering team as technical project lead to design and ship a full-stack marketplace web application.

RESEARCH & PUBLICATIONS

Thinking Inside the Box: Building UIs with Question-Guided Inputs | Graphics Interface Conference 2026
- Designed and built an interactive question-guided UI system for creating web applications; evaluated the interface through an HCI user study and published the work as a first-author paper at Graphics Interface 2026.

AWARDS

- Women in CS Graduate Scholarship ($3,000, 2026) · UWaterloo Graduate Scholarship ($4,000, 2025) · Mastercard Award in Cyber Security ($15,000, 2022) · Maurice William Summerhayes Memorial Scholarship ($40,000, 2020).